# Engineering Economy (INDE 301)

## Chapters 7 & 8 Homework — Problem Set

Rate of Return Analysis: Single & Multiple Alternatives

Instructor: Naim Al Khoury · Teaching Assistant: Mohammad Younis

Fall 26

---

**Instructions:**

- For every calculation, clearly state which interest factor(s) you are using in standard notation (e.g., $(P/A, i, n)$) before substituting numerical values.
- Show the formula you are using before computing the final numerical answer.
- Format all multi-step mathematical derivations on separate lines with clear alignment.
- Round final dollar answers to the nearest cent and interest rates to two decimal places, unless otherwise stated.

# Part A — Single-Alternative Rate of Return

## Exercise 1: Machinery Upgrade for a Manufacturing Facility

An industrial manufacturing facility is evaluating an energy-efficiency machinery upgrade over an 8-year service life. The initial capital expenditure is \$400,000, and the machinery generates a uniform net annual cost savings of \$75,000 per year for 8 years. The company's MARR is 12% per year.

1. Write the complete Present Worth equation defining the rate of return $i^*$, expressed in standard factor notation ($PW(i^*) = 0$).
2. Find the rate of return $i^*$ of this base project using linear interpolation.
3. State whether the base investment is economically acceptable under MARR = 12%.
4. To demonstrate the fundamental definition of ROR, construct the unrecovered balance schedule for Year 1 and Year 2 using $i^* = 10.01\%$. Verify that interest is charged strictly on the unrecovered balance rather than the original \$400,000.

In fact, the machinery can be resold at the end of Year 8 for a salvage value of \$60,000.

5. Re-write the updated Present Worth equation defining $i^*$.
6. Determine an initial estimate for the updated rate of return $i^*$ following the manual estimation procedure from the course notes, and find $i^*$ using linear interpolation.

7. Alternatively, explain how your previously calculated base rate ($i^* = 10.01\%$) serves as a direct shortcut to bracket the root without re-estimating from scratch.
8. Explain the economic impact of this salvage value on $i^*$, and state whether the investment becomes acceptable.

Suppose that upon further inspection, the machinery also requires a major mid-life component overhaul costing \$80,000 at the end of Year 5 (while retaining the \$60,000 salvage value at Year 8).

9. Re-write the updated Present Worth equation defining $i^*$ including the overhaul.
10. Determine an initial estimate for $i^*$ following the manual estimation procedure from the course notes from scratch (do not start from previous results). Using this estimate, bracket the root, calculate the final rate of return $i^*$ using linear interpolation, and verify whether the project remains acceptable under MARR = 12%.
11. The net cash flow in Year 5 is negative (−\$5,000). Evaluate the cash flow series using Descartes' rule of signs and Norstrom's criterion to prove whether a unique positive rate of return is mathematically guaranteed.

## Exercise 2: Rooftop Solar Array for a Regional Hospital

A regional hospital is evaluating a rooftop solar array with battery storage costing \$480,000 today. Net annual savings are \$62,000 at the end of Year 1 and increase by \$9,000 each year thereafter over a 10-year service life. At the end of Year 10, the equipment has a market value of \$40,000. The hospital's MARR is 12% per year.

1. Write the Present Worth equation as a function of $i^*$ in standard factor notation.
2. Determine an initial guess for $i^*$, and justify your choice of transformation.
3. Find $i^*$.
4. State whether the hospital should invest in the solar array based on your calculated $i^*$, and verify your decision by computing the Present Worth at the MARR.

## Exercise 3: Geothermal Power Station Acquisition

A geothermal power station requires an initial acquisition cost of \$850,000. It is projected to generate a net revenue of \$210,000 at the end of Year 1, but due to steam field pressure drop, revenues decline geometrically at a constant rate of $g = -7\%$ per year over its 9-year operational life with no salvage value. The MARR is 16% per year.

1. Write the Present Worth equivalence equation defining $i^*$ using standard geometric gradient factor notation $(P/A, g, i, n)$, and state the explicit analytical formula for the factor from the course notes.
2. Make an initial estimate for $i^*$ by ignoring the time value of money.
3. Calculate the internal rate of return $i^*$.
4. Determine if the buyer should accept the asking price of \$850,000 under MARR = 16%.

### Exercise 4: Equipment Purchase for a University Makerspace (Excel Analysis)

A university makerspace is offered a 6-year equipment purchase: pay \$250,000 now, then resell the equipment for \$60,000 at the end of Year 6. The MARR is 12% per year.

1. *Scenario 1:* The equipment saves a uniform \$55,000 at the end of each of the 6 years. State the exact Excel `RATE` function syntax to find $i^*$, calculate the rate, and state the investment decision.
2. *Scenario 2:* Savings are \$40,000 at the end of Year 1 and increase by \$6,000 each year thereafter. Set up the year-by-year cash flows in tabular form, state the exact Excel `IRR` function syntax, and decide.
3. Explain why the `RATE` function cannot be used in Scenario 2.

## Part B — Multiple Alternatives (Exercises 5–7)

### Exercise 5: Airport Shuttle Fleet for a Hotel Group

A hotel group is replacing the airport shuttle service at one of its resorts and has narrowed the choice to two fleets. Both fleets last 8 years, and the MARR is 10% per year. The electric fleet requires a replacement battery pack at the end of Year 4; the diesel fleet has no comparable mid-life expense.

| | Diesel fleet (A) | Electric fleet (B) |
|---|---|---|
| First cost | \$240,000 | \$410,000 |
| Net shuttle income | \$52,000 in Year 1, +\$4,000 each year | \$91,000 in Year 1, +\$4,000 each year |
| Battery-pack replacement | none | \$45,000 at the end of Year 4 |
| Salvage value (Year 8) | \$30,000 | \$60,000 |

1. Write the cash flow of each fleet year by year. Compute $i^*$ for each fleet and screen them against the MARR. Is each fleet viable on its own?

2. Develop the incremental cash flows $(B - A)$. Explain why the \$4,000 annual gradient cancels out completely.
3. Estimate the incremental rate of return $\Delta i^*_{B-A}$ using linear interpolation.
4. Which fleet should the hotel group choose? The diesel fleet has a higher individual $i^*$, so why is it not selected?
5. The battery supplier warns that the pack may cost more than \$45,000 by Year 4. Determine the maximum replacement price at Year 4 for which the electric fleet remains economically justified over the diesel fleet.

## Exercise 6: Gaming Rigs for a University Esports Arena

A university esports arena rents out gaming stations for tournaments and open-play hours. It must choose between two hardware plans. Whichever plan is chosen is renewed with identical cash flows when it wears out, and the MARR is 12% per year.

| | Option A: mid-range PCs (life 2 yr) | Option B: high-end rigs (life 3 yr) |
|---|---|---|
| First cost | \$48,000 | \$85,000 |
| Net rental revenue | \$24,000 in Year 1, +\$2,000 in Year 2 | \$30,000 in Year 1, +\$2,000 each year |
| Salvage value (end of life) | \$9,000 | \$14,000 |

1. Write the cash flow of each option over its own life cycle.
2. Compute $i^*$ for each option and screen them against the MARR.
3. The two options have different useful lives. Determine the study period needed to compare them using the Present Worth method.
4. Tabulate the extended cash flows over the study period and determine the incremental cash flow series $(B - A)$.
5. Estimate the incremental rate of return $\Delta i^*_{B-A}$ using linear interpolation.
6. Which option should the arena select?
7. Verify the selection using the Annual Worth (AW) method on a single life cycle of each option. Explain why no LCM is needed when using the AW method.

### Exercise 7: Automation Packages for a Distribution Center

A distribution center will install *exactly one* of four automation packages. Each package lasts 6 years, and the MARR is 12% per year. The packages differ in first cost and in the annual savings they deliver.

| Package | First cost | Net annual savings | Salvage (Year 6) |
|---|---|---|---|
| A: Conveyor upgrade | \$120,000 | \$40,000 per year | \$10,000 |
| B: Pick-to-light system | \$210,000 | \$45,000 per year | \$15,000 |
| C: Sorting robots | \$260,000 | \$75,000 per year | \$25,000 |
| D: Automated storage | \$380,000 | \$100,000 per year | \$40,000 |

1. Compute $i^*$ for each package and screen against the MARR. Which package has the highest individual $i^*$, and which package is eliminated?
2. List the surviving packages in order of increasing first cost and conduct a systematic pairwise incremental ROR analysis.
3. Which package should be selected?
4. A junior analyst compares Package D directly with Package A, finds $\Delta i^*_{D-A} \geq 12\%$, and recommends Package D. Compute $\Delta i^*_{D-A}$ and explain why this comparison is fundamentally flawed.
5. Verify the selection by computing the Present Worth of each package at MARR = 12%.

## Part C — Conceptual Multiple-Choice Questions

**1.** The rate of return $i^*$ of a project is best defined as:

(a) the interest rate applied to the original investment for the whole project life;

(b) the interest rate at which the present worth of the net cash flows equals zero;

(c) the minimum attractive rate of return the company would like to earn;

(d) the arithmetic average of the yearly percentage returns.

**2.** In an unrecovered-balance table for a project that earns exactly $i^*$, the interest charged in Year $t$ is computed on:

(a) the original investment;

(b) the cash flow received in Year $t$;

(c) the average of the beginning and ending balances;

(d) the unrecovered balance at the beginning of Year $t$.

**3.** A net cash flow has 4 sign changes. Descartes' rule of signs states that the number of real rates of return with $i > -100\%$ is:

(a) at most 4;

(b) exactly 4;

(c) exactly 1;

(d) at least 1.

**4.** According to Norstrom's criterion, a cash flow series has a unique positive rate of return when:

(a) the raw cash flows start positive and never change sign;

(b) the raw cash flows change sign at least twice;

(c) the cumulative cash flows start negative and change sign exactly once;

(d) the cumulative cash flows end at zero.

**5.** A project has two rates of return, $-12\%$ and $+18\%$, and the MARR is 10%. By the course guidelines for multiple RORs, the analyst should:

(a) discard both roots;

(b) use $+18\%$ as the ROR and accept the project;

(c) use $-12\%$ as the ROR and reject the project;

(d) average the two roots.

**6.** A project costs \$60,000 and returns \$24,000 at the end of each of 3 years (no salvage value). Its rate of return is closest to:

(a) 6.67%;

(b) 40.00%;

(c) 0.00%;

(d) 9.70%.

**7.** Mutually exclusive revenue alternatives A (first cost \$50,000, $i^* = 22\%$) and B (first cost \$120,000, $i^* = 17\%$) are compared at a MARR of 10%. The incremental rate of return is $\Delta i^*_{B-A} = 12\%$. Which should be selected?

(a) A, because it has the highest individual $i^*$;

(b) Neither, because $\Delta i^*$ is below both individual $i^*$ values;

(c) B, because the extra investment earns at least the MARR;

(d) Do nothing, because $\Delta i^*$ exceeds the MARR.

**8.** Which step of the incremental ROR procedure is *skipped* when all mutually exclusive alternatives are cost alternatives?

(a) Computing the individual $i^*$ of each alternative;

(b) Ranking the alternatives by first cost;

(c) Computing the incremental cash flows;

(d) Comparing $\Delta i^*$ with the MARR.

**9.** Two mutually exclusive revenue alternatives have unequal lives (3 and 4 years). When the incremental ROR is evaluated with the AW method, the analyst should:

(a) extend both alternatives to a 12-year LCM before computing AW;

(b) use only the shorter life for both alternatives;

(c) compute the PW over the longer life and ignore the shorter one;

(d) set $AW_B(i) = AW_A(i)$ using the single life cycle of each alternative.

**10.** Excel's `RATE(n, A, P, F)` function can be used directly to find the rate of return only when the cash flow consists of:

(a) any arbitrary pattern of annual net cash flows;

(b) one initial amount, a uniform annual series, and one final amount;

(c) a uniform series plus an arithmetic gradient;

(d) several sign changes.